DTFT and its properties

Practice questions · Day 06 · Thu 29 Oct · pack D3

5. Practice

From your course files

T4 Q1 (tutorial). Compute the Fourier transform of the following signals.

(a) $x(n) = u(n) - u(n-6)$

(b) $x(n) = 2^n u(-n)$

(c) $x(n) = \left(\frac{1}{3}\right)^n u(n+4)$

(d) $x(n) = \{-2, -1, \underset{\uparrow}{0}, 1, 2\}$

T4 Q2 (tutorial). Consider the signal

$$x(n) = \{-1, 2, \underset{\uparrow}{-3}, 2, -1\}$$

with Fourier transform $X(\omega)$. Compute the following quantities without explicitly computing $X(\omega)$:

(a) $X(0)$

(b) $\angle X(0)$

(c) $\int_{-\pi}^{\pi} X(\omega)\, d\omega$

(d) $X(\pi)$

(e) $\int_{-\pi}^{\pi} |X(\omega)|^2\, d\omega$

T4 Q3 (tutorial). The following input-output pairs have been observed during the operation of various LTI systems. Determine the frequency response of each system.

(a) $x(n) = \left(\frac{1}{2}\right)^n u(n) \xrightarrow{T_1} y(n) = \left(\frac{1}{8}\right)^n u(n)$

(b) $x(n) = e^{j\frac{\pi}{5}n} \xrightarrow{T_2} y(n) = 3e^{j\left(\frac{\pi}{5}n - \frac{\pi}{3}\right)}$

(c) $x(n) = u(n) - u(n-1) \xrightarrow{T_3} y(n) = \left(\frac{1}{5}\right)^{n+2} u(n+2)$

T4 Q4 (tutorial). Consider an FIR system described by the difference equation

$$y(n) = x(n) + x(n-4).$$

(a) Compute and sketch the magnitude and phase response of this system.
(b) Determine its response to the input signals:
(i) $x(n) = \cos\left(\frac{\pi}{4}n\right) + 3\sin\left(\frac{\pi}{3}n + \frac{\pi}{10}\right)$
(ii) $x(n) = 10 + 5\cos\left(\frac{2\pi}{5}n + \frac{\pi}{2}\right)$

New questions

N1 [3 marks] (easy). $x(n) = \{1, \underset{\uparrow}{1}, 1\}$.

a. Find $X(\omega)$ in real form. [2]
b. Evaluate $X(0)$ and $X(\pi)$. [1]

N2 [4 marks]. $x(n) = (-0.5)^n u(n+1)$.

a. Find $X(\omega)$. [2]
b. Find $X(0)$ and $X(\pi)$ using the shortcuts, and check them against (a). [1]
c. Is $x(n)$ a lowpass or a highpass signal? [1]

N3 [5 marks]. $x(n) = \{2,\ -1,\ \underset{\uparrow}{3},\ 0,\ 1\}$ has DTFT $X(\omega)$. Without computing $X(\omega)$, find:

a. $X(0)$ (b) $X(\pi)$ (c) $\int_{-\pi}^{\pi} X(\omega)d\omega$ (d) $\int_{-\pi}^{\pi}|X(\omega)|^2 d\omega$ (e) $\int_{-\pi}^{\pi} X(\omega)e^{j2\omega}d\omega$ (f) $\left.\frac{dX}{d\omega}\right|_{\omega=0}$ [5 total]

N4 [6 marks] (multi-part, chained). Let $x(n) = (0.5)^n u(n)$ and $y(n) = x(n) * x(n)$.

a. Find $Y(\omega)$. [1]
b. Use differentiation in frequency to show $y(n) = (n+1)(0.5)^n u(n)$. [2]
c. Find $Y(0)$ and $Y(\pi)$, and check $Y(0)$ from the time domain. [1]
d. Let $v(n) = y_{(2)}(n)$ (time expansion by 2). Find $V(\omega)$, $V(0)$ and $V(\pi)$. [2]

N5 [5 marks] (sampling). $x_a(t) = \cos(2\pi\, 150t) + 0.5\cos(2\pi\, 400t)$ is sampled at $F_s = 500$ Hz.

a. Find the digital frequencies of both tones in $(-\pi, \pi]$. [2]
b. Sketch or describe $X(\omega)$ on $(-\pi, \pi]$. [1]
c. What does an ideal lowpass reconstruction filter (gain T, cutoff 250 Hz) output? [1]
d. What is the smallest F_s that avoids aliasing? [1]

N6 [5 marks] (hard: modulation and windowing). Let $x(n) = u(n) - u(n-6)$ (the pulse in the figure) and $y(n) = x(n)\cos(\pi n)$.

a. Use the modulation property to write $Y(\omega)$ in terms of $X(\omega)$. [2]
b. Find $Y(0)$ and $Y(\pi)$. [1]
c. Find $\int_{-\pi}^{\pi}|Y(\omega)|^2 d\omega$. [1]

d. In one sentence, describe $|Y(\omega)|$ compared with the figure. [1]

6. Quick recall check

1. Write $X(\pi)$ for $x(n) = \{4, \underset{\uparrow}{1}, 2\}$.
2. $x(n) \leftrightarrow X(\omega)$. What is the DTFT of $x(-n-3)$?
3. A 1.2 kHz tone is sampled at 2 kHz. What is the digital frequency in $(-\pi, \pi]$, and what is the apparent analogue frequency?
4. For the DTFS, which side carries the $1/N$, and what does c_0 represent?
5. $x(n)$ is real and odd. What can you say about $X(\omega)$?